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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 13

Modulation and Demodulation

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COMPUTER · CHAPTER 13

Modulation and Demodulation

ਮੋਡੂਲੇਸ਼ਨ ਅਤੇ ਡੀਮੋਡੂਲੇਸ਼ਨ

IN THIS CHAPTER

- Modulation And Demodulation
- Modulation
- Demodulation
- Modulation Vs Demodulation
- Types Of Modulation
- Frequency Bands

MODULATION AND DEMODULATION

Communication systems transmit information over a communication medium using signals. Before transmission, data must be converted into a suitable signal format.

MODULATION

Modulation is the process of converting information (data) into a suitable analog signal by combining it with a carrier signal for transmission through a communication medium.

Modulation means **adding information to a carrier signal** so that it can travel over long distances.

Example

When you speak on a mobile phone, your voice is converted into electrical signals and modulated before transmission.

DEMODULATION

Demodulation is the process of extracting the original information from the received modulated signal.

Demodulation means recovering the original data from the received signal.

Example

When the receiver hears your voice during a phone call, the received signal is demodulated back into sound.

MODULATION VS DEMODULATION

Feature	Modulation	Demodulation
Meaning	Adding information to carrier signal	Extracting information from signal
Location	Sender Side	Receiver Side
Purpose	Transmission	Recovery of Data

MODEM (MOdulator DEModulator)

A **Modem** is a communication device that performs both:

- Modulation
- Demodulation

CARRIER SIGNAL

A **Carrier Signal** is a signal that contains no information but has characteristics that can carry information.

Characteristics of Carrier Signal

1. Amplitude

2. Frequency

3. Phase

MODULATED SIGNAL

A **Modulated Signal** is formed by combining:

- Carrier Signal
- Information (Message) Signal

Formula

Modulated Signal = Carrier Signal + Information Signal

Exam Point

Modulated Signal carries actual information.

CHARACTERISTICS OF A SIGNAL

A signal is mainly characterized by:

Characteristic	Meaning	Unit
Amplitude	Strength of Signal	Volt (V)
Frequency	Number of Cycles per Second	Hertz (Hz)
Phase	Position of Wave	Degree (°) or Radian

1. Amplitude

Amplitude is the maximum height or strength of a signal.

Unit :Volt (V)

Example :Higher amplitude means stronger signal strength.

2. Frequency

Frequency is the number of cycles completed by a signal in one second.

Unit :Hertz (Hz)

Example :1 Hz = 1 cycle per second

3. Phase

Phase represents the position of a wave relative to a reference point.

Unit :Degree (°) or Radian

Exam Point

Amplitude → Volt

Frequency → Hertz

Phase → Degree (°) or Radian

TYPES OF MODULATION

Information can be added to a carrier signal by changing:

- Amplitude
- Frequency
- Phase

1. AMPLITUDE MODULATION (AM)

In Amplitude Modulation, the amplitude of the carrier signal changes while frequency and phase remain constant.

Example :AM Radio Broadcasting

2. FREQUENCY MODULATION (FM)

In Frequency Modulation, the frequency of the carrier signal changes while amplitude remains constant.

Example :FM Radio Broadcasting

3. PHASE MODULATION (PM)

In Phase Modulation, the phase of the carrier signal changes according to the information signal.

Characteristics

- Better signal quality
- Better noise resistance
- Higher efficiency

Phase Modulation is generally more efficient than AM and FM.

FREQUENCY BANDS

Frequency bands are ranges of frequencies used for communication.

1. RADIO FREQUENCY (RF) BAND

Radio Frequency Band is used for wireless transmission of signals and data.

Uses

- Mobile Communication
- Wi-Fi
- Radio Broadcasting
- Television Broadcasting

2. VOICE BAND

Voice Band is the frequency range used for telephone voice communication.

Uses

- Telephone Calls
- Voice Communication Systems

3. NARROWBAND

A Narrowband communication channel uses a small bandwidth.

Example

Traditional Telephone Networks

Exam Point

Narrowband = Low Bandwidth

4. BROADBAND

Broadband communication uses a wide frequency range and supports high-speed data transmission.

Examples

- Fiber Internet
- Broadband Internet Connections
- Cable Internet

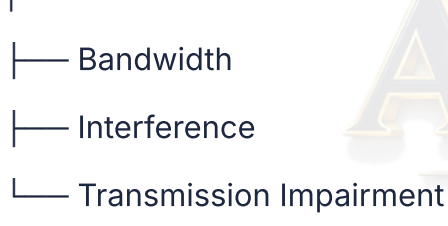
Exam Point

Broadband = High Bandwidth + High Speed

FACTORS AFFECTING TRANSMISSION MEDIA

The efficiency of a communication channel depends on several factors.

Transmission Media Factors



BANDWIDTH

Bandwidth is the maximum amount of data that can be transmitted through a communication channel in a given period of time.

Units of Bandwidth

Unit	Full Form
bps	Bits per Second
Kbps	Kilobits per Second
Mbps	Megabits per Second

Unit	Full Form
Gbps	Gigabits per Second
Tbps	Terabits per Second

Example

A 100 Mbps Internet connection can transfer more data than a 10 Mbps connection.

Memory Trick

Bandwidth = Road Width

Wider Road → More Vehicles

Higher Bandwidth → More Data

INTERFERENCE

Interference is the disturbance of a signal caused by unwanted external signals.

Effects

- Data Errors
- Signal Quality Reduction
- Communication Failure

Example

Noise from electrical equipment affecting radio communication.

TRANSMISSION IMPAIRMENT

Transmission Impairment occurs when the received signal differs from the transmitted signal.

Result

- Poor signal quality
- Data corruption
- Communication errors

CAUSES OF TRANSMISSION IMPAIRMENT

Transmission Impairment

│

├── Attenuation

├── Distortion

└── Noise

1. ATTENUATION

Attenuation means loss of signal strength as the signal travels through a transmission medium.

Cause

Signal loses energy while travelling long distances.

Solution

Amplifiers are used to restore signal strength.

Example

Mobile signals become weaker at longer distances.

2. DISTORTION

Distortion is a change in the shape or form of a signal during transmission.

Cause

Different frequency components travel at different speeds and reach the destination at different times.

Example

Poor-quality audio transmission.

3. NOISE

Noise is an unwanted signal that mixes with the original signal.

Types of Noise

1. Thermal Noise

Produced due to heat in electronic devices.

2. Crosstalk Noise

Occurs when signals from one channel interfere with another channel.

3. Induced Noise

Caused by external electromagnetic sources.

4. Impulse Noise

Sudden spikes in voltage caused by lightning or power disturbances.

MOST IMPORTANT FACTS FOR EXAMS

1. Modem stands for Modulator-Demodulator.

2. Carrier signal contains no information.

3. Amplitude is measured in Volts.

4. Frequency is measured in Hertz.

5. Phase is measured in degrees or radians.

6. AM changes amplitude.

7. FM changes frequency.

8. PM changes phase.

9. Phase Modulation is more efficient than AM and FM.

10. RF Band is used for wireless communication.

11. Broadband provides high-speed data transfer.

12. Bandwidth is measured in bps.

13. Attenuation means signal loss.

14. Distortion means signal shape change.

15. Noise means unwanted signal.

16. Amplifiers compensate for attenuation